

Helium

He (He I). Strongest classified transitions below 2 μm (NIST): 53 levels, 79 lines.

He⁺ ionisation limit 198310.67 cm⁻¹ = 24.58739 eV (50.426 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle f||er||f' \rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

